Chapter 12

Inlays, onlays and veneers

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INTRODUCTION

Historically, gold was the material of choice for inlays and onlays. The accuracy and malleability of gold made it an ideal material to restore lost tooth structure and protect weakened cusps destroyed by caries and trauma. Although gold is versatile, its use has declined over recent years but is still used by those clinicians who recognize its unique qualities in areas of the mouth where appearance is not important. Modern attitudes to appearance and colour have seen a preference for tooth-coloured materials.

Modern materials utilize the properties of adhesive cements to reduce the necessity for extensive tooth preparation and restoration, so conserving tooth tissue. Both ceramic and composite-based materials are regularly used as an alternative to gold in restoring broken down teeth, and veneers have been designed to improve the appearance of minimally restored teeth.

INLAYS AND ONLAYS

The most common indication for inlays or onlays is extensively restored and weakened teeth. The distinction between the two designs is unclear but simply those restorations without cuspal coverage that remain within the body of the tooth would be considered inlays (Figure 12.1) whereas onlays replace tooth tissue including cusps (Figure 12.2).

Inlays

Inlays are usually used when difficulty is anticipated or experienced in obtaining an acceptable contour, contact point and occlusion on a directly placed restoration. Repeated fracture of a directly placed restoration may also indicate the placement of an inlay, and in this situation a gold inlay may be advocated for its strength. Composite and ceramic inlays may also be chosen for their aesthetics in situations where strength is not a major requisite. This can be particularly relevant for the mesial surfaces of upper premolar teeth and the occlusal surfaces of lower teeth when appearance is important to the patient. Increasingly, patients are concerned about the placement of amalgam and request an alternative when the restorations need replacement and in rare situations amalgam replacement is necessary because of lichenoid reactions.

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Figure 12.1 Gold inlay placed due to repeated fracture of the previous amalgam restorations. Aesthetics is not a problem in this upper premolar tooth.



Figure 12.2 Gold mesial occlusal distal (MOD) with cuspal overlay; the tooth preparation can be seen on the left with the fit surface of the restoration to the right.

In these situations, directly placed composites must be considered the most appropriate material to restore minimal cavities. The necessity of using an indirect material with associated elimination of undercuts and laboratory costs means their indications are limited to more extensive cavities.

Onlays

The most commonly placed partial coverage extracoronal restoration would be an onlay where weakened tooth structure can be protected without further extensive tooth removal. A common indication for an onlay would be a root-filled posterior tooth where cuspal protection is required. Root canal treatment in posterior teeth is usually the result of caries and restorative procedures and, as such, these teeth are extensively broken down and have weakened cusps. The coronal access cavity needed for a root treatment removes the roof of the pulp chamber, weakening the tooth further, and can leave a limited amount of buccal and lingual tooth tissue which might be completely removed if prepared for a crown (Figure 12.3). Preservation of some part of the buccal and lingual tooth helps to retain the core and reduces the need to consider a post, especially in premolar teeth (Figure 12.4). The three-quarter gold crown preparation on a premolar tooth, particularly when only the buccal cusp remains, will preserve tooth tissue but the colour and appearance of gold within the smile line is considered by some to be a disadvantage (see Chapter 8 and Figure 12.5).



Figure 12.3 The upper left first premolar tooth has an extensive mesial-occlusal-distal (MOD) amalgam restoration; preparation of this tooth for a metal–ceramic crown might result in the loss of the entire buccal cusp, leaving the amalgam unsupported or poorly retained, with potential loss of the restoration. An onlay or inlay preparation would preserve the buccal tissue.



Figure 12.4 Onlays cemented to both premolars restoring teeth which were extensively destroyed by caries. By using an onlay rather than a crown the buccal tooth tissue is preserved, avoiding the need for a full coverage restoration.



Figure 12.5 Three-quarter crown preparation on the lower left second premolar tooth has preserved buccal tooth tissue, but may be regarded by some patients as having an unacceptable appearance.

Bonding materials (including gold) to a tooth, using adhesive cements, reduces some of the need for conventional principles of retention. Onlays can be considered when there is no or little intracoronal shape to the preparation and where retention is poor. Despite the retention provided by adhesive cement, conventional concepts of tooth preparation for retention should not be ignored and, where possible, should be incorporated into the design of the preparation (Figures 12.2 and 12.6); routine use of adhesive cements to achieve retention poses problems if retrieval is ever necessary.

Dental materials

Gold

Whilst gold is gradually being superseded by tooth-coloured materials for the provision of indirect restorations, its use should not be ignored as gold has some advantages over indirect composites and ceramics when used for inlays and onlays. It has high strength, ductility and has the ability to be cast accurately, especially into thin sections. A dentist would be unable to obtain full consent from a patient if gold (with its advantages and disadvantages) was not discussed as a treatment option.



Figure 12.6 The intracoronal preparation for an onlay. The sharp line angles on the buccal and lingual surfaces allow the porcelain or indirect composite an ideal margin. Internal line angles can be rounded.

Gold onlays need less occlusal and axial reduction than that needed for tooth-coloured materials, as the latter tend to be more brittle and require greater bulk for strength. The ductility of gold and its ability to be highly polished means that it is not as abrasive to the opposing dentition, especially for those patients with tooth wear. All intra- and extra-coronal restorations made from gold need tooth preparation with near parallel walls and an absence of undercuts (Figure 12.7).

Composite

The need for tooth-coloured inlays competes with directly placed composites for aesthetically pleasing tooth-coloured restorations. Laboratory-made indirect composites have a theoretical reduction in polymerization shrinkage as the curing is completed in the laboratory. The curing occurs by subjecting the composite to heat, pressure or intense light, and this higher degree of cure improves strength and reduces wear; the indirect composite inlays/onlays are often referred to as 'super cured'. This super curing does, however, leave fewer reactive resin groups to bond to the resin luting cement. The accuracy of fit is significantly less than restorations made in either ceramic or gold and they can absorb dietary stains after a few years which significantly affects their appearance.

Ceramic

Ceramic restorations are indicated where the aesthetic demands of the patient are high. The colour matching of modern porcelains allows the margin between the tooth and material to be almost imperceptible. Where there is a slight difference in colour, the margin of a ceramic inlay/onlay will be obvious (see Figure 12.4) as there is a sudden transition from ceramic to tooth. Ceramic materials, unlike directly placed composite, cannot be placed into progressively thinner layers to blend and merge the



Figure 12.7 Disto-occlusal inlay preparation and cemented inlay. The retentive features in the inlay preparation, i.e. near parallel walls and occlusal key, are adequate and adhesive resin luting cements are not necessary.

colour to that of the tooth, due to their brittleness. This progressive transition with direct composite can camouflage the restoration. Ceramic is a hard material and has high wear resistance; however, it can be abrasive to the opposing dentition when used on the occlusal surfaces, especially if the glaze is lost and the restoration has been adjusted.

The manufacturing techniques used for porcelain and indirect composites generally cannot tolerate such parallel preparations as those produced for a gold inlay, and more tapered walls are needed. Undercuts in the tooth preparation can either be eliminated by further tooth preparation or be blocked out with an adhesive tooth-coloured material. All variants of the preparations should avoid sharp line angles, particularly intracoronally (Figure 12.8).

The traditional method of manufacture of onlays and inlays was the lost wax technique for gold-based materials and the use of refractory dies for all-ceramic restorations (see manufacture of veneer later in this chapter). More recently, computer-aided design and computer-aided manufacture (CAD-CAM) technology has allowed laboratory and chairside production of indirect restorations. Most systems utilize scanners to record the tooth preparation or its model digitally and then send the data to an automated milling machine. The milled crown, inlay or onlay has a fit accuracy of a few microns and is adjusted to fit the occlusion with the opposing tooth. There are a variety of commercial systems on the market and all use the same basic principles (see Chapter 11).



Figure 12.8 Tooth prepared for ceramic inlay. Note that undercuts are blocked out with bonded composite, avoiding further tooth removal to eliminate undercuts. The preparation is slightly more tapered and has rounded internal line angles. *(Courtesy of Dr Suzanne Blacker)*

Ceramic inlays and onlays are cemented with a resin-based luting cement and are handled in the same way as described for ceramic veneers below.

TOOTH PREPARATION FOR INLAYS AND ONLAYS

Tooth preparation for intra- and extra-coronal restorations follows the same basic concepts as used for all indirect restorations. The preparation should avoid undercuts between opposing walls within the cavity; however, the more complex the shape, the harder it is to avoid them. Multiple surface intracoronal restorations, particularly those including proximal boxes, are particularly demanding and, as a result, cavities tend to be over-tapered. Since all-ceramic restorations rely upon the cement lute for most of the retention, provided there are no undercuts a slightly over-tapered cavity is acceptable. However, gold restorations gain most of their retention from the cavity shape and are therefore more reliant upon near parallel preparations.

The major advantage for inlay and onlay restorations is the preservation of some tooth tissue to retain the core. If existing cavities contain undercuts they can be blocked out with composite or glass ionomer cements to provide the necessary cavity shape (Figures 12.9 and 12.10). Tapered burs provide the most convenient shape to prepare inlays and onlays and reduce the chance of creating undercuts. If an onlay preparation is to be cut, occlusal clearance/reduction will be required consistent with the material chosen. Similarly, the marginal configuration (shoulder, chamfer or deep chamfer) will also be dictated by the material planned.

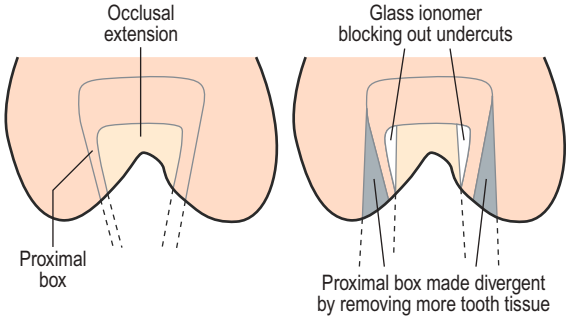


Figure 12.9 Diagrammatic representation of an undercut Class II cavity which has been prepared for an amalgam restoration (left). To convert this to an inlay preparation, the undercuts have to be eliminated. This can be achieved either by blocking out the undercuts with an adhesive restorative material or by removing more tooth tissue to produce a divergent cavity (right).



Figure 12.10 The upper left first molar tooth has been prepared for an inlay which is to act as a retainer for a cantilever bridge. A silver reinforced glass ionomer has been used to block out the undercut and all margins should be on tooth tissue. *Note*: Further removal of the material is required at the floor of the box to allow the margin of the inlay to be placed on tooth tissue.

CERAMIC VENEERS

Ceramic veneers are thin layers of ceramic (Figure 12.11) used to mask discoloured teeth and to alter the shape and alignment of minimally restored teeth. It is important to understand that whilst tooth preparation for veneers is relatively conservative of tooth tissue compared to full coverage crowns, some tooth preparation is required and the procedure is irreversible. Adequate tooth preparation should allow for a veneer 0.5–0.75 mm thick.

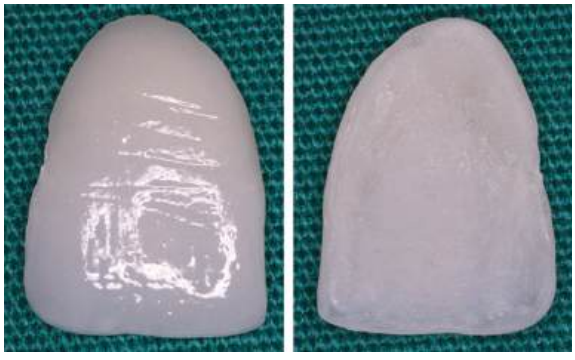


Figure 12.11 Labial (left) and fit (right) surfaces of a ceramic veneer.

Specific considerations

If poorly aligned in-standing teeth are veneered, they may not need as much reduction, as the veneer thickness can be used to align the teeth (Figure 12.12). However, the outer, aprismatic, surface of mature enamel should always be removed as it provides an inferior bond compared to that of etched subsurface prismatic enamel. When using ceramic veneers to improve the appearance of restored teeth, small cavities containing old discoloured composite restorations should be replaced to minimize the appearance of any shine-through and to ensure a superior bond from the veneer to the newly placed composite. Water sorption and limited reactive chemical groups found in old polymerized composites lead to an inferior bond with the luting resin, leakage, staining and increased failure (Figure 12.13). When treatment planning for veneers, careful attention to the occlusion between the restoration and the opposing tooth is needed as those placed under heavy occlusal loads are more likely to debond or fracture. In the upper anterior region, for example, caution is needed when veneers are planned for teeth with incisal edge-to-edge contacts, cross bites and signs of parafunctional habits such as wear facets (Figure 12.14).

Tooth preparation for veneers

When discoloured teeth are well aligned, a uniform reduction in the labial enamel should be carried out (Figure 12.15). Inadequate tooth reduction leads to the veneer appearing too bulky or a veneer too thin to mask the discolouration. Uniform reduction of tooth tissue can be difficult when done 'freehand and eyeballed'. Multiple dimples cut with a round diamond bur of radius ~0.5 mm and held at 45° to the tooth surface may help as a depth guide. The angulation



Figure 12.12 Ceramic veneers have been used to align the incisors in a patient with a Class II Division II incisor relationship, with retroclined central incisors. Note the contact points are broad bucco-lingually.



Figure 12.13 The ceramic veneer on the upper right central incisor has been placed on an old distal composite. An inferior bond has led to leakage and staining.



Figure 12.14 Inappropriate placement of a ceramic veneer on the upper right lateral incisor tooth, which is in cross bite with the lower canine, has led to fracture and partial debond of the veneer.

that the bur is held ensures that the dimples do not become too deep as the shank of the bur engages the tooth surface (Figure 12.16). A specially made bur with a blank shank and diamond wheels (Figure 12.17) can also be used in a similar manner to make horizontal depth cuts in the enamel. A tapered bur can then be used to reduce the remaining labial tooth tissue until the dimples or horizontal grooves just disappear. With all burs it is important to remember that the labial face of a tooth is curved and the



Figure 12.15 Discoloured upper right central incisor where a ceramic veneer is planned. It is in alignment with the upper left central incisor and the labial face will have to be reduced to create space for the veneer.



Figure 12.16 Dimples cut into the labial face with the bur at 45° to the tooth ensures the dimples do not become too deep. The dimples act as depth cuts, guiding uniform labial reduction for the veneer.

reduction needs to conform to this (Figure 12.18) as with any crown preparation.

At the cervical margin a chamfer finish should be created with a round-ended tapered bur. Some burs have a finer diamond grit at the tip to ensure better adaptation and fit of the ceramic veneer at the margin (Figure 12.19). The veneer preparation should be extended proximally, sufficient to completely mask the tooth tissue and prevent a dark line being seen at the margins (Figure 12.20). However, the contact point between teeth must be preserved, if temporary veneers are not placed, as proximal space closure will prevent the veneer from seating fully.

Incisally, a number of preparations have been described, each with their advantages and disadvantages (Figure 12.21). The feather edge finish is easy to prepare



Figure 12.17 Bur with smooth shank and diamond wheels used for producing horizontal depth cut grooves to aid veneer preparation.

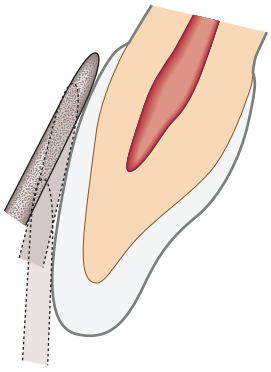


Figure 12.18 The tapered diamond bur should be held in at least three different planes to ensure the labial curvature of the preparation conforms to that of the tooth, giving an even 0.5 mm reduction.



Figure 12.19 Tapered bur with finer diamond grit tip to produce a chamfer margin with fewer irregularities.



Figure 12.20 Inadequate extension of the veneer preparations proximally has led to the appearance of dark proximal lines.

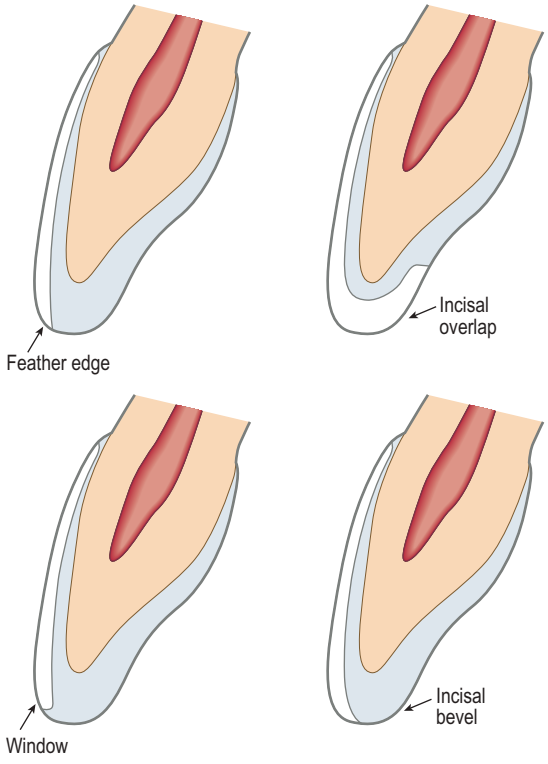


Figure 12.21 Various types of incisal preparation that have been described for veneers.

and there is little risk of damage to the ceramic in most occlusal schemes; however, it does not allow a translucent incisal tip on the ceramic (Figure 12.22). The incisal overlap overcomes this but the preparation is more destructive and the palatal finish line is in a position where occlusal contacts could lead to damage. The window preparation is difficult to prepare and is not normally recommended; however, the incisal margin of the veneer is protected from occlusal



Figure 12.22 Feather edge finish to the incisal aspect of the veneer preparation has led to no incisal translucency and a monotone finish. Excess luting cement still has to be removed.

forces. Unfortunately, the incisal edge of the tooth may be visible, compromising the result (Figure 12.23). The incisal bevel overcomes this but the ceramic margin becomes thin and fragile and is more suited for restorations using a directly placed composite veneer. The choice of preparation will depend upon each clinical situation, but those most commonly used are the feather edge or incisal overlap. Figure 12.24 shows the completed veneer preparation for the patient seen in Figures 12.15 and 12.16.

Temporization

Once the tooth preparation has been completed, an impression is taken and consideration can be given to temporization. Placement of temporary veneers is usually unnecessary, particularly if a conservative intra-enamel preparation has been carried out. Preparing teeth for a veneer reduces the thickness of enamel and if the teeth are markedly discoloured the appearance may become worse. If temporary veneers are considered necessary, these are best made out of directly placed composite resin and bonded to the tooth by acid etching a small spot of the enamel, leaving the majority of the enamel unaffected and ready for etching for the definitive veneers. Composite veneers 'spot welded' in such a way are easy to remove.



Figure 12.23 Window type preparation for these composite veneers has led to a darker incisal line.



Figure 12.24 Veneer preparation for the patient seen in Figures 12.15 and 12.16 completed until no signs of dimples remain. The margins are prepared to a chamfer finish cervically and proximally, and a feather edge incisally.

Manufacture of ceramic veneers

Two models are required when making ceramic veneers. The first model is made from a refractory die material onto which the ceramic can be built and which can withstand the temperature needed within the furnace when the ceramic is fired (Figure 12.25). A glaze porcelain is usually applied to the refractory die before building up the ceramic. The ceramic

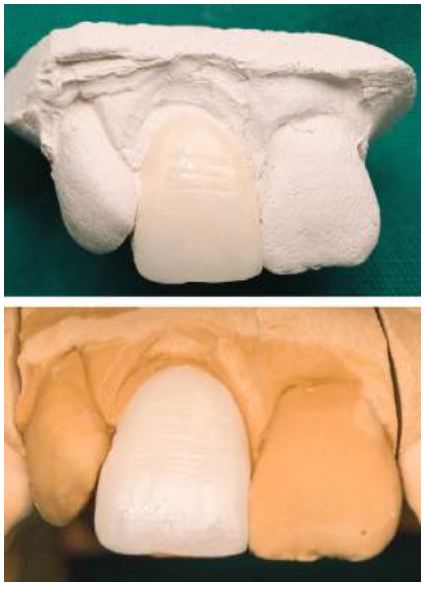


Figure 12.25 Refractory die (above) onto which the ceramic is built and fired. Once the refractory die has been sandblasted from the fit surface of the ceramic it can be tried on the stone master die (below).

powder is usually made from feldspar (80%), quartz (15%) and kaolin (5%) which is mixed with an aqueous medium to produce a paste. This is built up on the refractory die and the particles are condensed by vibration and water absorption using a tissue paper. The refractory die and ceramic are then put through a firing cycle which fuses the porcelain particles together. Once removed from the furnace and cooled, the refractory die material is sandblasted from the veneer, taking care not to damage the thin laminate. This also roughens the fit surface of the veneer, improving the bond with the luting cement. The fit surface may be roughened further by etching with hydrofluoric acid gel. This is extremely caustic and should not come into contact with patients; its use is confined to the laboratory. The second model made by the dental technician is a conventional die stone model onto which the veneer is seated to assess fit and shape.

As with all-ceramic crowns, ceramic veneers can also be produced using CAD-CAM technology (see Chapter 11).

Try-in and cementation

To try-in and cement a veneer, an appropriate resin luting cement system should be used. These usually have water soluble try-in pastes of various shades which correspond to the definitive resin luting cement and can modify the overall shade of the veneer. Try-in pastes are generally composed of either glycerine or polyethylene glycol resin to which has been added a variety of additives and pigments. They exhibit the phenomenon of shear thinning which allows the controlled loading of the fit surface of the veneer, with no running off of the material. The subsequent reduction in viscosity that occurs upon seating the restoration upon the tooth both facilitates positioning and minimizes the chances of porcelain fracture occurring in this procedure. Thereafter the viscosity increases, minimizing the chances of displacement of the veneer by the patient's circumoral musculature whilst the appearance is assessed. Even if the ceramic appears to be a good colour match, a neutral or clear try-in paste should be used as this can alter the light transmission and reflection at the veneer–tooth interface which can affect the appearance (Figure 12.26).

When operator and patient are happy with the appearance of the veneer at try-in, cementation can take place. The water soluble try-in paste should be thoroughly rinsed off and the veneer dried. A silane coupling agent is then applied to the fit surface of the veneer and gently air dried. This chemical exhibits good wetting but degrades in the presence of both moisture and air. Its application should therefore be as near as possible to the cementation of the restoration. The silane is a chemical coupling agent that will bond the porcelain to the luting agent.

For cementation the tooth should ideally be isolated under a rubber dam and a split-dam technique may avoid interference with the seating of the veneer cervically (Figure 12.27). A cellulose strip should be placed between



Figure 12.27 Teeth isolated using a split rubber dam technique, in preparation for cementation of veneers.

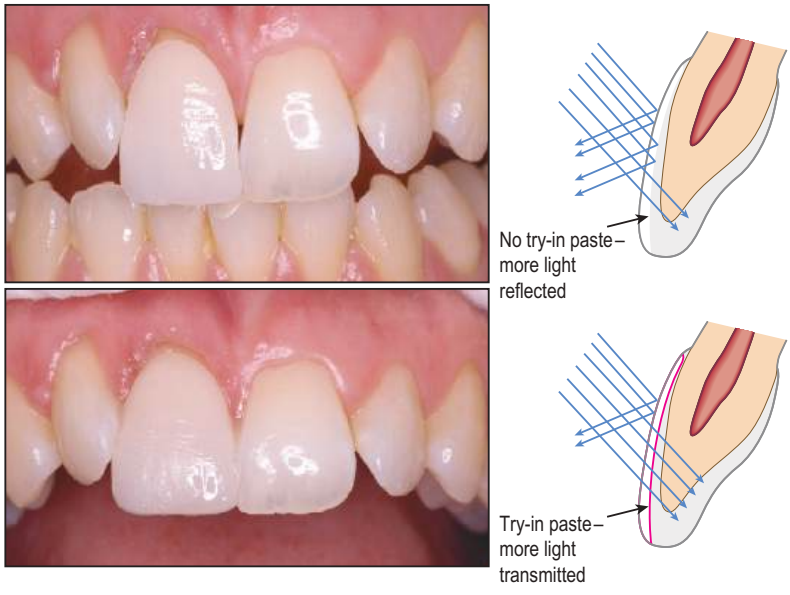


Figure 12.26 Ceramic veneer try-in; with no try-in paste above and with a neutral try-in paste below. Note that the veneer looks lighter when no try-in paste is used due to greater reflection of light from the veneer surface and the veneer tooth interface – placement of a try-in paste and the definitive luting cement lead to more light transmission.

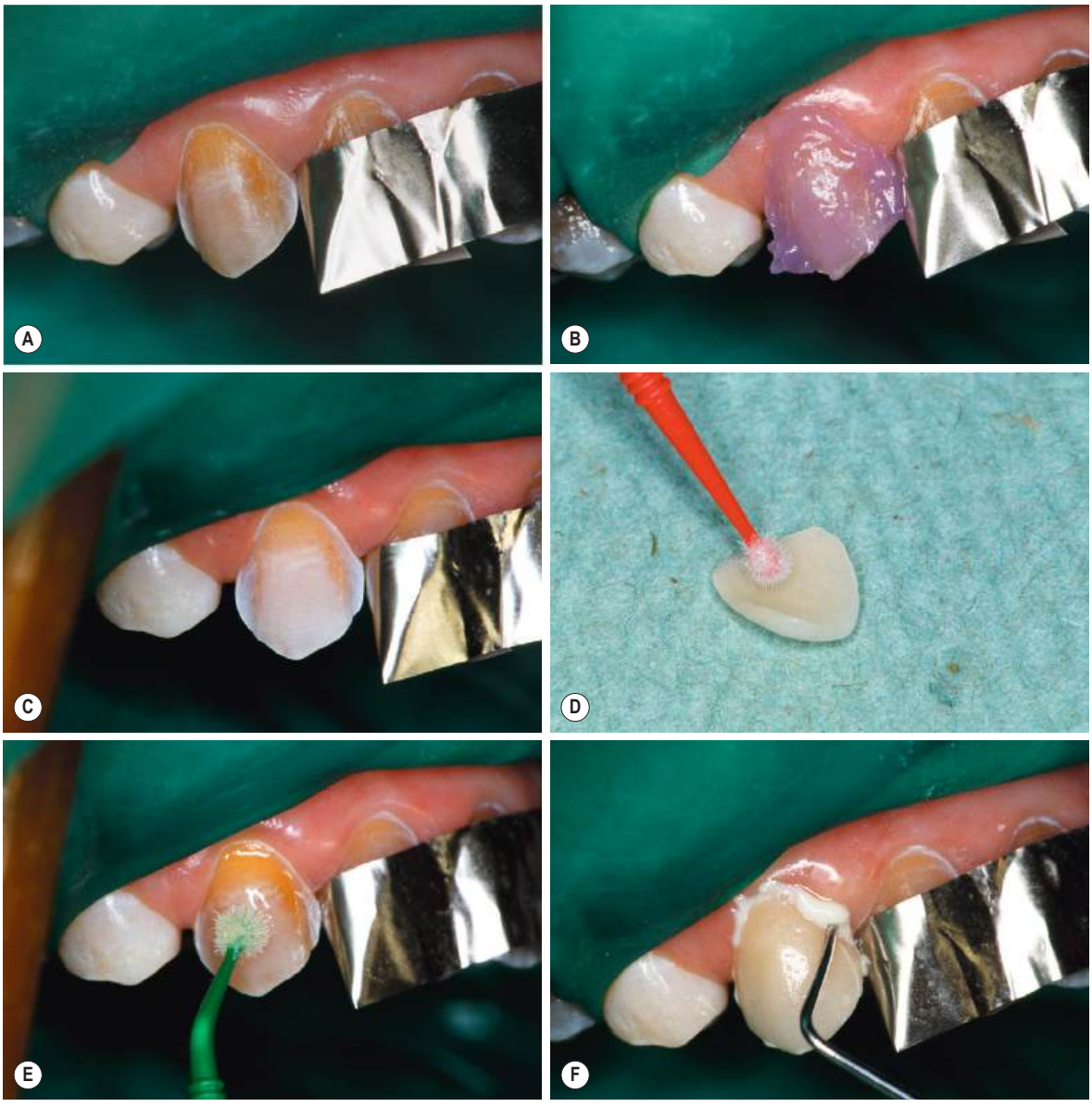


Figure 12.28 Stages in cementing a ceramic veneer. Tooth isolation (**A**), acid etch tooth (**B**), rinse and dry thoroughly (**C**), apply silane coupling agent to veneer (**D**), apply bond to tooth (**E**), load the veneer with resin luting cement and seat, removing excess before light curing (**F**).

the teeth to avoid bonding adjacent teeth together. If difficulty is experienced with access to the tooth due to the cellulose strip, a soft, thin, malleable metal strip can be used instead which can be moulded around the contact point of the adjacent teeth, giving better access (Figure 12.28).

Once the prepared tooth has been acid etched, rinsed, dried, bond applied and cured, the resin luting cement can be loaded onto the fit surface of the veneer and seated (Figure 12.28). Excess cement can be removed with a microbrush dipped in resin and the cement light cured, as most resin luting cements used for cementing veneers are dual cured. It is better to leave a little excess luting cement which can be trimmed away and polished, rather than overzealously trying to remove all excess which can lead to cement lute being dragged out from the veneer–tooth interface.

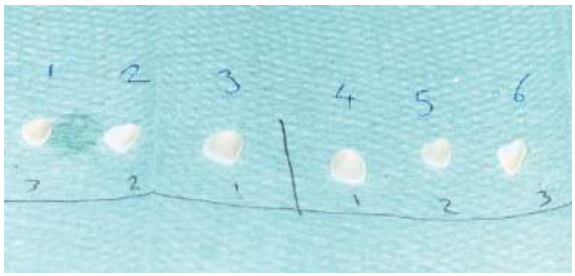


Figure 12.29 Veneers for patient seen in Figure 12.27. The order in which the veneers were tried-in was from right to left. It is important to repeat the same order when cementing so as not to alter the paths of insertion.

When cementing multiple veneers, each veneer should be tried-in individually to assess fit and appearance. The veneers then need to be tried-in *en masse* and the order in which they are seated should be recorded (Figure 12.29). This order should be repeated when they are finally cemented to avoid problems that can arise with different paths of insertion when a different order of cementation is used.



Figure 12.30 This patient has tooth wear affecting the palatal surfaces of the upper anterior teeth. The incisors are short and require crowns; the canines are only affected palatally and require gold palatal veneers.

GOLD VENEERS

Tooth wear can either affect the entire dentition or can be localized, typically to the upper anterior teeth (see Chapter 4). The pattern of tooth tissue loss can also vary. Consider the patient in Figure 12.30: all of the upper anterior teeth are worn palatally, but only the incisors have lost crown height and require crowns. The palatal wear on the canine teeth can simply be restored with gold alloy veneers, the tooth wear having effectively prepared the teeth so that no further preparation is required (Figure 12.31). Originally high copper gold alloys were oxidized or tin plated to achieve improved bond strengths with adhesive resin luting cements such as Panavia (Kuraray Dental). However, recent data (Chana et al., 2000) have shown that sandblasting gold veneers with alumina and cementation with Panavia 21 leads to clinically acceptable survival rates.

Where tooth wear occurs labially as well as palatally, some clinicians have suggested 'double veneers', that is, gold palatal veneers and ceramic labial veneers. Although this might be modestly more conservative of tooth tissue, the multiple interfaces between veneers and tooth tissue make these restorations less predictable than metal–ceramic crowns and the latter are usually favoured in such a clinical scenario.



Figure 12.31 Patient seen in Figure 12.30, with metal–ceramic crowns placed on the incisor teeth and palatal gold veneers on the canine teeth.

SUMMARY

Inlays and onlays often provide a conservative alternative to full coverage crown preparations for the restoration of heavily restored teeth which are either not suitable for direct restorations or where direct restorations have failed. Similarly, ceramic veneers can alter the appearance of teeth, avoiding the need for more destructive all-ceramic or metal–ceramic crowns. However, the merits of directly placed composites should not be overlooked.

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